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CHEMISTRY

Paper 1 Multiple Choice

Grade 12

May 2016

1 hour

Candidates answer on the Question Paper.

Additional Materials: Data Booklet
 Electronic calculator

12CHEM/01

READ THESE INSTRUCTIONS FIRST

Write your Centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct by putting a tick (✓) in the correct box.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working can be done in this booklet.

Electronic calculators may be used.

For Examiner's Use

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This document consists of **20** printed pages.

Section AFor
Examiner's
Use

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct by putting a tick (✓) in the correct box.

- 1 Which statement is true about a zirconium ion, ${}_{40}^{90}\text{Zr}^{3+}$?

- A** There are 37 electrons in this ion.
B There are 43 electrons in this ion.
C There are 50 nucleons in this ion.
D There are 50 protons in this ion.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 2 Use of the Data Booklet is relevant to this question.

A piece of sodium of mass 0.460 g reacts with an excess of chlorine gas.

How many chlorine molecules, Cl_2 , react with the sodium?

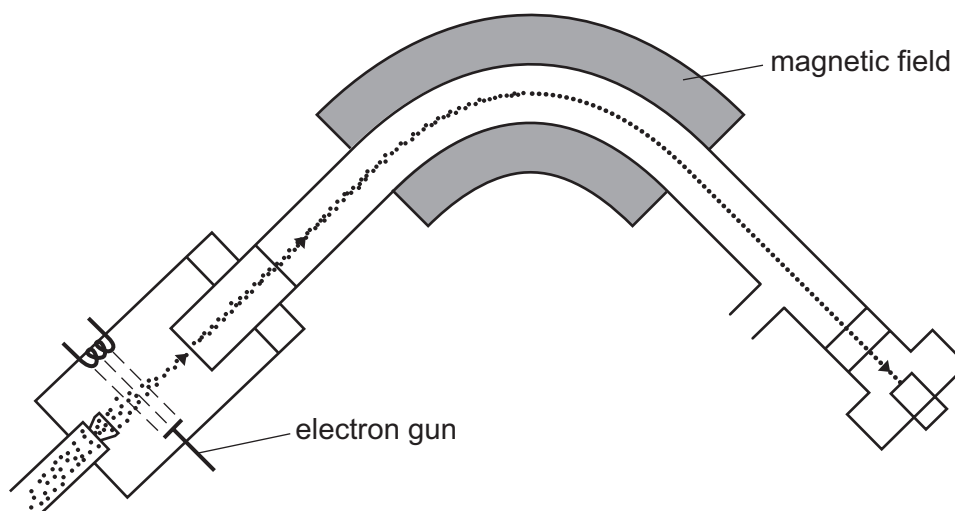
- A** 6.020×10^{21}
B 1.204×10^{22}
C 2.408×10^{22}
D 6.020×10^{23}

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 3 The diagram shows a simplified mass spectrometer.

For
Examiner's
Use



Which row of the table correctly describes the actions of the electron gun and the magnetic field?

	action of the electron gun	action of the magnetic field
A	acceleration	deflection
B	acceleration	detection
C	ionisation	deflection
D	ionisation	detection

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 4 How many different absorptions are there in the C-13 NMR spectrum of 3-ethylpentane?

- A** 1
B 3
C 5
D 7

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 5 The first ionisation energy of magnesium ($\Delta H_{\text{ie}}^{\circ} \text{Mg}$) is greater than the first ionisation energies of sodium ($\Delta H_{\text{ie}}^{\circ} \text{Na}$) and calcium ($\Delta H_{\text{ie}}^{\circ} \text{Ca}$).

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Which factors can be used to explain these differences?

	$\Delta H_{\text{ie}}^{\circ} \text{Mg} > \Delta H_{\text{ie}}^{\circ} \text{Na}$	$\Delta H_{\text{ie}}^{\circ} \text{Mg} > \Delta H_{\text{ie}}^{\circ} \text{Ca}$
A	nuclear charge	nuclear charge
B	nuclear charge	shielding
C	shielding	nuclear charge
D	shielding	shielding

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 6 Methane, CH_4 , has four pairs of electrons around the central carbon atom. Methane has a bond angle of 109.5° . Ammonia, NH_3 , also has four pairs of electrons around the central atom. However, ammonia has a bond angle of 107° .

Which statement explains this difference in bond angle?

- A** Ammonia molecules can hydrogen bond with each other.
B Lone pair to bond pair repulsion is greater than bond pair to bond pair repulsion.
C Nitrogen atoms are more electronegative than carbon atoms.
D The hydrogen atoms in CH_4 repel each other more than the hydrogen atoms in NH_3 .

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

7 Use of the Data Booklet is relevant to this question.

For
Examiner's
Use

The reactivity of different metals as reducing agents can be compared using data.

Which row of the table is correct?

	comparing cobalt and manganese as reducing agents	comparing chromium and vanadium as reducing agents
A	cobalt is stronger than manganese	chromium is stronger than vanadium
B	cobalt is stronger than manganese	vanadium is stronger than chromium
C	manganese is stronger than cobalt	chromium is stronger than vanadium
D	manganese is stronger than cobalt	vanadium is stronger than chromium

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

8 Which row of the table is correct?

	strength as reducing agents	explanation
A	Cl^- ions > Br^- ions > I^- ions	I^- ions can reduce Cl^- ions in aqueous solution
B	Cl^- ions > Br^- ions > I^- ions	larger anions lose electrons more easily than smaller anions
C	I^- ions > Br^- ions > Cl^- ions	I^- ions can reduce Cl^- ions in aqueous solution
D	I^- ions > Br^- ions > Cl^- ions	larger anions lose electrons more easily than smaller anions

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 9 The hydroxide compounds of the Group 2 elements become more soluble down Group 2.

Which statement explains this?

- A** $\Delta H^\circ_{\text{hyd}}$ becomes less exothermic down Group 2;
 $\Delta H^\circ_{\text{lat}}$ becomes more exothermic down Group 2.
- B** $\Delta H^\circ_{\text{hyd}}$ becomes more exothermic down Group 2;
 $\Delta H^\circ_{\text{lat}}$ becomes less exothermic down Group 2.
- C** $\Delta H^\circ_{\text{hyd}}$ and $\Delta H^\circ_{\text{lat}}$ both become less exothermic down Group 2;
 $\Delta H^\circ_{\text{hyd}}$ changes more than $\Delta H^\circ_{\text{lat}}$.
- D** $\Delta H^\circ_{\text{hyd}}$ and $\Delta H^\circ_{\text{lat}}$ both become less exothermic down Group 2;
 $\Delta H^\circ_{\text{lat}}$ changes more than $\Delta H^\circ_{\text{hyd}}$.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 10 An acidified solution of dichromate(VI) ions, $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$, will oxidise $\text{Sn}^{2+}(\text{aq})$ ions to $\text{Sn}^{4+}(\text{aq})$ ions. Chromium is reduced to $\text{Cr}^{3+}(\text{aq})$ in the reaction.

What is the ionic equation for this redox reaction?

- A** $\text{Cr}_2\text{O}_7^{2-} + \text{Sn}^{2+} + 14\text{H}^+ \rightarrow \text{Cr}^{3+} + \text{Sn}^{4+} + 7\text{H}_2\text{O}$
- B** $\text{Cr}_2\text{O}_7^{2-} + 3\text{Sn}^{2+} + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{Sn}^{4+} + 7\text{H}_2\text{O}$
- C** $\text{Cr}_2\text{O}_7^{2-} + 6\text{Sn}^{2+} + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 6\text{Sn}^{4+} + 7\text{H}_2\text{O}$
- D** $2\text{Cr}_2\text{O}_7^{2-} + 3\text{Sn}^{2+} + 28\text{H}^+ \rightarrow 4\text{Cr}^{3+} + 3\text{Sn}^{4+} + 14\text{H}_2\text{O}$

A ☐ **B** ☐ **C** ☐ **D** ☐

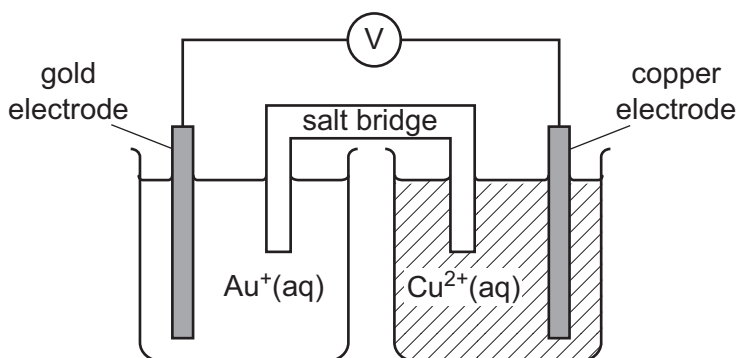
[1]

For
Examiner's
Use

11 Use of the Data Booklet is relevant to this question.

For
Examiner's
Use

The electrochemical cell shown is set up under standard conditions.



The voltmeter reading is 1.34 V. Cu/Cu^{2+} is the negative half-cell of the electrochemical cell.

What is the standard electrode potential, $E^\ominus$, for Au^+/Au half-cell?

- A – 1.00 V
- B – 0.50 V
- C + 0.84 V
- D + 1.68 V

A ☐ B ☐ C ☐ D ☐

[1]

12 Copper forms a complex with ethane-1,2-diamine, $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$. The formula of the complex is $[\text{Cu}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$. In this complex, a total of six lone pairs of electrons from the ligands bond with the copper ion. These six lone pairs are arranged octahedrally around the copper ion.

What can be deduced from this information alone?

- A Ethane-1,2-diamine is a bidentate ligand.
- B The $[\text{Cu}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$ complex is insoluble in water.
- C The $[\text{Cu}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$ complex is colourless in aqueous solution.
- D The $[\text{Cu}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$ complex is less stable than the $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ complex.

A ☐ B ☐ C ☐ D ☐

[1]

- 13 The temperature used in the Haber process is 450 °C.

Which statement describes why this temperature is used?

- A 450 °C gives an acceptable yield of ammonia at an acceptable rate.
- B 450 °C gives the maximum possible rate of reaction.
- C 450 °C gives the maximum possible yield of ammonia at equilibrium.
- D 450 °C keeps energy costs as low as possible.

A ☐ B ☐ C ☐ D ☐

[1]

- 14 The four statements **all** correctly describe the use of a catalyst.

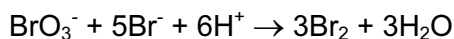
Which of the catalysts is a **homogeneous** catalyst here?

- A Divanadium pentoxide is the catalyst in the Contact process.
- B Fe^{2+} ions catalyse the reaction between I^- ions and $\text{S}_2\text{O}_8^{2-}$ ions in aqueous solution.
- C Iron is the catalyst in the Haber process.
- D Nickel catalyses the reaction between ethene and hydrogen.

A ☐ B ☐ C ☐ D ☐

[1]

- 15 Bromate ions, BrO_3^- , will oxidise bromide ions in acidic solution.



The reaction is first order with respect to BrO_3^- , first order with respect to Br^- and second order with respect to H^+ . The rate of the reaction is $6.60 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$ when $[\text{BrO}_3^-]$ is $0.0200 \text{ mol dm}^{-3}$, $[\text{Br}^-]$ is $0.0100 \text{ mol dm}^{-3}$ and $[\text{H}^+]$ is $0.500 \text{ mol dm}^{-3}$.

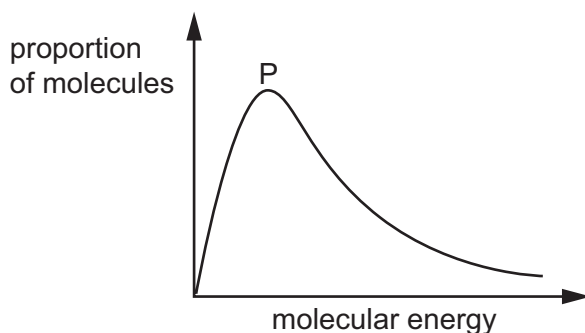
What is the value of the rate constant, k ?

- A 6.60×10^{-5}
 B 0.660
 C 1.32
 D 33.0

A ☐ B ☐ C ☐ D ☐

[1]

- 16 The Maxwell-Boltzmann distribution curve for the energies of the molecules in a sample of gas at temperature 300 K is shown.



Which statement about peak P is correct?

- A At 290 K peak P is to the left of and lower than its position at 300 K.
 B At 290 K peak P is to the right of and lower than its position at 300 K.
 C At 310 K peak P is to the left of and higher than its position at 300 K.
 D At 310 K peak P is to the right of and lower than its position at 300 K.

A ☐ B ☐ C ☐ D ☐

[1]

17 What is the definition of the *enthalpy change* of a chemical reaction?

- A The amount of energy equal to $\Delta G - T\Delta S$ for this reaction.
- B The amount of energy needed to break the bonds in the reaction.
- C The heat evolved or absorbed when the reaction is carried out at constant pressure.
- D The heat evolved or absorbed when the reaction is carried out at constant volume.

A ☐ B ☐ C ☐ D ☐

[1]

18 Which equation correctly represents a step in the photochemical chlorination of ethane?

- A $\text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_5\cdot + \text{H}\cdot$
- B $\text{C}_2\text{H}_6 + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{H}\cdot$
- C $\text{C}_2\text{H}_5\text{Cl} + \text{H}\cdot \rightarrow \text{C}_2\text{H}_4\text{Cl}\cdot + \text{H}_2$
- D $\text{C}_2\text{H}_6 + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\cdot + \text{HCl}$

A ☐ B ☐ C ☐ D ☐

[1]

- 19 A large sample of the compound $\text{CH}_3\text{CHCHCH}_3$ is tested with a small amount of bromine water.

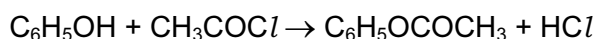
Which row of the table correctly describes the result of this test and the conclusion that can be drawn?

	result of the test	conclusion
A	the bromine water is decolourised	$\text{CH}_3\text{CHCHCH}_3$ is a saturated compound
B	the bromine water is decolourised	$\text{CH}_3\text{CHCHCH}_3$ is an unsaturated compound
C	the bromine water remains brown	$\text{CH}_3\text{CHCHCH}_3$ is a saturated compound
D	the bromine water remains brown	$\text{CH}_3\text{CHCHCH}_3$ is an unsaturated compound

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 20 Phenol reacts with ethanoyl chloride. An acylation reaction takes place.



What is the first step in the mechanism of this reaction?

- A** Attack by an electrophile on an electron deficient carbon atom.
B Attack by a nucleophile on an electron deficient carbon atom.
C Homolytic fission of an oxygen-hydrogen bond.
D Homolytic fission of a carbon-chlorine bond.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

21 Which name and structure are correct?

- A $\text{CH}_3\text{CH}_2\text{CHO}$ is propanone
 B $\text{C}_2\text{H}_5\text{COC}_2\text{H}_5$ is butanone
 C $(\text{CH}_3)_2\text{CHCH}_2\text{CHO}$ is 3-methylbutanal
 D $\text{C}_2\text{H}_5\text{COCH}_3$ is butan-3-al

A ☐ B ☐ C ☐ D ☐

[1]

22 Each of the carbon atoms in one molecule of $\text{CH}_2\text{CHCHCHCH}_2\text{COOH}$ is bonded to several neighbouring atoms. The angles made by the bonds around each of the carbon atoms in the molecule are close to either 109.5° or 120° .

How many of each type of carbon atom are there in one molecule of this compound?

	carbon atoms with 109.5° bond angles	carbon atoms with 120° bond angles
A	1	5
B	2	4
C	4	2
D	5	1

A ☐ B ☐ C ☐ D ☐

[1]

23 Which statement correctly describes a **two-step** synthesis which could be performed by making nitrobenzene in the first step?

- A making aspirin from benzene
 B making chlorobenzene from benzene
 C making phenol from benzene
 D making phenylamine from benzene

A ☐ B ☐ C ☐ D ☐

[1]

- 24 Nitrobenzene can be made by reacting benzene with a nitrating mixture. A nitrating mixture is made by mixing concentrated nitric acid with concentrated sulfuric acid. This mixture produces the ion **X** that attacks benzene, resulting in the formation of nitrobenzene.

What is **X**?

- A NO_2^-
B NO_3^-
C NO_2^+
D NO_3^+

A ☐ B ☐ C ☐ D ☐

[1]

- 25 Phenylmethylketone, $\text{C}_6\text{H}_5\text{COCH}_3$, is a chemical with many uses, including making fragrances.

Phenylmethylketone can be made from benzene and ethanoyl chloride by a Friedel-Crafts reaction. An aluminium chloride catalyst is used. The mechanism of this reaction involves a positively charged intermediate that contains a six-membered ring.

What is the molecular formula of this intermediate?

- A $\text{C}_7\text{H}_9\text{O}^+$
B $\text{C}_8\text{H}_8\text{O}^+$
C $\text{C}_8\text{H}_9\text{O}^+$
D $\text{C}_8\text{H}_9\text{OCl}^+$

A ☐ B ☐ C ☐ D ☐

[1]

- 26 Compound **W** is treated with hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$. The product mixture is found to contain an ester with molecular formula $\text{C}_6\text{H}_{12}\text{O}_2$. This substance is different from **W**.

What could **W** be?

- A cyclohexan-1,4-diol
- B ethan-1,2-diol
- C hexan-2-ol
- D propan-1-ol

A ☐ B ☐ C ☐ D ☐

[1]

- 27 Stereoisomerism should be taken into account when answering this question.

How many isomeric products are formed when butan-2-ol is treated with an excess of concentrated sulfuric acid?

- A 2
- B 3
- C 4
- D 5

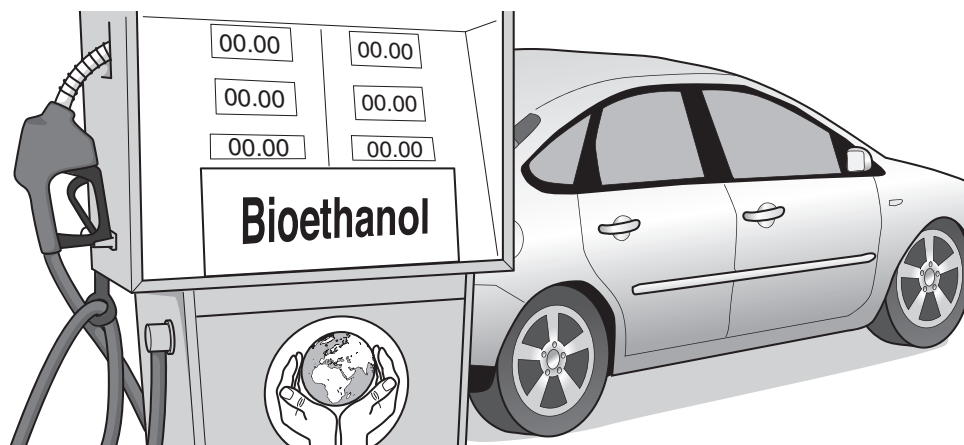
A ☐ B ☐ C ☐ D ☐

[1]

28 Use of the Data Booklet is relevant to this question.

For
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This fuel pump dispenses bioethanol. The ethanol has been made by fermentation of plant waste.



A factory makes 1260 tonnes of glucose, $C_6H_{12}O_6$, from plant waste.

What is the **theoretical maximum** mass of ethanol that could be made from this mass of glucose by fermentation?

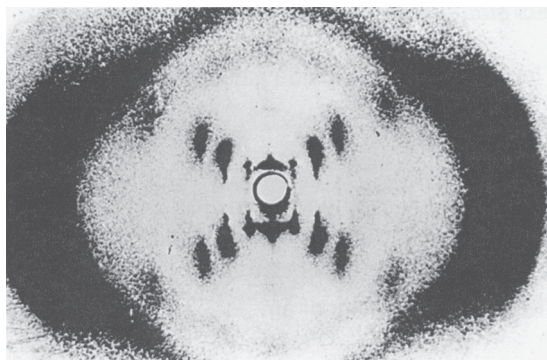
- A 322 tonnes
- B 644 tonnes
- C 852 tonnes
- D 1260 tonnes

A ☐ B ☐ C ☐ D ☐

[1]

- 29 DNA consists of two polymer strands wound round each other in a double helix. Important evidence for this double helix was obtained by Rosalind Franklin and Raymond Gosling, using X-ray crystallography.

For
Examiner's
Use



Franklin and Gosling's X-ray crystallography image obtained from DNA

The sub-units of DNA are described as sugar, phosphate and four types of organic base. The sugar and the phosphate form the backbone of each DNA polymer strand.

Which description of the location of the organic bases is correct?

- A on the inside of the double helix, joined to each sugar unit
- B on the inside of the double helix, joined to each phosphate unit
- C on the outside of the double helix, joined to each sugar unit
- D on the outside of the double helix, joined to each phosphate unit

A ☐ B ☐ C ☐ D ☐

[1]

- 30 The structural formula of aspartic acid is $\text{HOOCCH}_2\text{CH}(\text{NH}_2)\text{COOH}$.

A sample of aspartic acid is stored at pH 1. The electric charge on each aspartic acid molecule is X.

Another sample of aspartic acid is stored at pH 14. The electric charge on each aspartic acid molecule is Y.

What is the value of X minus Y?

- A +1
- B +2
- C +3
- D +4

A ☐ B ☐ C ☐ D ☐

[1]

Section BFor
Examiner's
Use

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

31 Which statement(s) help to explain why increased levels of carbon dioxide gas in the atmosphere might cause global warming?

- 1 Carbon dioxide molecules can absorb infrared radiation.
- 2 The C=O bonds in a carbon dioxide molecule are polar.
- 3 The density of carbon dioxide is greater than the average density of air.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

32 Which statement(s) is/are correct?

- 1 Bromine is more electropositive than iodine.
- 2 Sodium is more electropositive than magnesium.
- 3 Carbon is more electronegative than silicon.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

33 Which statement(s) is/are correct?

- 1 5.145 g of NaBr ($M_r=102.9$) dissolved in 100 cm³ of solution has a concentration of 0.500 mol dm⁻³.
- 2 34.0 g of NH₃ ($M_r=17.0$) dissolved in 500 cm³ of solution has a concentration of 4.00 mol dm⁻³.
- 3 7.98 g of CuSO₄ ($M_r=159.6$) dissolved in 250 cm³ of solution has a concentration of 0.200 mol dm⁻³.

A ☐ B ☐ C ☐ D ☐

[1]

34 The statements that follow refer to the benefits and risks of using a hydrogen-oxygen fuel cell as the energy source in an electric car.

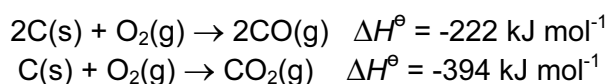
Which statement(s) is/are correct?

- 1 The product of the reaction in the fuel cell is water.
- 2 If the fuel tank splits in an accident there will be no risk to the driver.
- 3 Hydrogen is a renewable fuel which can be made easily by fermenting sugars.

A ☐ B ☐ C ☐ D ☐

[1]

35 The standard enthalpy changes for two reactions are given.



Which statement(s) is/are correct?

- 1 The standard enthalpy change of formation of carbon monoxide is -222 kJ mol⁻¹.
- 2 The reaction $\text{CO}_2\text{(g)} + \text{C(s)} \rightarrow 2\text{CO(g)}$ is endothermic.
- 3 The standard enthalpy change of formation of carbon dioxide is -394 kJ mol⁻¹.

A ☐ B ☐ C ☐ D ☐

[1]

- 36 A sample of ethanol vapour occupies 200 cm^3 at 800K . It is completely combusted in excess oxygen. The volumes of water vapour and carbon dioxide produced are measured at 800K .

Assume that all reactants and products behave as ideal gases at this temperature.

Which statement(s) is/are correct?

- 1 The volume of carbon dioxide produced is 400 cm^3 .
- 2 The volume of water vapour produced is 600 cm^3 .
- 3 The volume of oxygen consumed is 700 cm^3 .

A ☐ B ☐ C ☐ D ☐

[1]

- 37 Which statement(s) about optical isomers and optical isomerism is/are correct?

- 1 A pure optical isomer will rotate the plane of polarised light.
- 2 A medical drug and its optical isomer may have different effects on a patient.
- 3 A mixture of a compound and its optical isomer is called a racemic mixture.

A ☐ B ☐ C ☐ D ☐

[1]

- 38 The sequence of atoms $-\text{O}(\text{CH}_2)_5\text{COO}(\text{CH}_2)_5\text{COO}(\text{CH}_2)_5\text{CO}-$ repeats down the whole length of a polymer chain.

Which monomer or monomers could be used to make this polymer?

- 1 $\text{HOOC}(\text{CH}_2)_5\text{COOH}$ and $\text{HOCH}_2(\text{CH}_2)_3\text{CH}_2\text{OH}$
- 2 $\text{HOCH}_2(\text{CH}_2)_4\text{COOH}$
- 3 $\text{HOCH}_2(\text{CH}_2)_4\text{COC}l$

A ☐ B ☐ C ☐ D ☐

[1]

39 **Z** is a compound of carbon, hydrogen and oxygen only. Separate samples of **Z** were tested and the following observations were recorded.

- When sodium metal was added to **Z**, effervescence was seen.
- When hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$ was added to **Z**, no colour change was seen.
- When **Z** was warmed with Tollens' reagent, no silver mirror was seen.

What could **Z** be?

- 1 $(\text{CH}_3)_3\text{COH}$
- 2 CH_3COOH
- 3 CH_3COCOOH

A ☐ B ☐ C ☐ D ☐

[1]

40 Use of the Data Booklet is relevant to this question.

Which amino acids have a non-polar R group?

- 1 leucine
- 2 asparagine
- 3 threonine

A ☐ B ☐ C ☐ D ☐

[1]

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